Anjun Chu

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Education

- **University of Colorado Boulder** — **Boulder, USA**
  PhD in Physics — *Aug 2018 – Aug 2024*
  - GPA: 4.0/4.0
  - PhD Thesis: *New Directions in Quantum Simulation and Metrology via Contact and Photon-mediated Interactions*
  - Thesis Supervisor: Prof. Ana Maria Rey
- **Tsinghua University** — **Beijing, China**
  BSc in Physics — *Aug 2014 – Jul 2018*
  - GPA: 91/100
  - Bachelor Thesis: *Magnetic Field Control of Sodium Spinor BEC*
  - Thesis Supervisor: Prof. Luming Duan

Research Experience

- **University of Chicago, Pritzker School of Molecular Engineering** — **Chicago, USA**
  Chicago prize postdoctoral fellow in theoretical quantum science — *Sep 2024 – Present*
  I currently focus on theory of quantum metrology, simulation and computation. In collaboration with Prof. Aashish Clerk, I worked on reservoir engineering protocols and their applications in quantum metrology and many-body state engineering.
- **University of Colorado Boulder, JILA** — **Boulder, USA**
  Graduate research assistant, supervised by Prof. Ana Maria Rey — *May 2019 – Aug 2024*
  I worked on the theory of quantum many-body dynamics in spin systems and their multilevel extensions, as well as their applications in quantum simulation and quantum metrology, in close collaboration with experimental groups of optical lattice clocks (Prof. Jun Ye) and cavity QED systems (Prof. James Thompson).
- **Tsinghua University, Center for Quantum Information** — **Beijing, China**
  Undergraduate researcher, supervised by Prof. Luming Duan — *May 2016 – Jul 2018*
  I worked on both theory and experiment of spinor BEC system. I participated in the construction of the sodium spinor BEC apparatus, the numerical simulation and the experimental measurement of dynamical quantum phase transition in this system.

Awards and Honors

- **Chicago Prize Postdoctoral Fellowship in Theoretical Quantum Science** *2024-Present*
  Awarded to exceptional researchers working in all areas of theoretical quantum science and provide recipients with the flexibility to collaborate with quantum theory faculty in University of Chicago
- **Boeing Quantum Creators Prize** *2023*
  Awarded to early-career researchers in the broad areas of quantum science and engineering whose work moves quantum science in new directions, selected by Chicago Quantum Exchange
- **Tsinghua Academic Excellence Scholarship** *2016,2017*
  Awarded to top 5 of 100 students of each department
- **Tsinghua Xuetang Scholarship** *2015-2018*
  Awarded to top students selected to Tsinghua Xuetang Talent Program for conducting undergraduate research

Service

- Research article reviewer for *Physical Review Letters*, *Physical Review Research*, *Physical Review A*, *Physical Review B*, *npj Quantum Information*
- Funding proposal reviewer for Gordon and Betty Moore Foundation
- Session chair for APS DAMOP Meeting, FINESS workshop

Publications

* These authors contributed equally to this work.

29. *Experimental observation of Higgs oscillations and their power-law decay across pairing regimes in a cavity QED quantum simulator*
Chitose Maruko, Haoqing Zhang, Eliot A. Bohr, Leah Huzjak, **Anjun Chu**, Victor Gurarie, Chengyi Luo, Ana Maria Rey, James K. Thompson
arXiv.2610.14171 (2026)

28. *Pump-Free Microwave-Optical Bell Pair Generation for Teleportation-Based Quantum Transduction*
Fangxin Li, Jaesung Heo, Zhaoyou Wang, Benjamin Pingault, Xingyu Gao, Tengyang Ruan, **Anjun Chu**, David D. Awschalom, Andrew N. Cleland, Andrew P. Higginbotham, Alexander A. High, Liang Jiang
arXiv:2609.16364 (2026)

27. *Collective three-body interactions enable a robust quantum speedup*
Haoqing Zhang, **Anjun Chu**, Chengyi Luo, Chitose Maruko, Eliot A. Bohr, James K. Thompson, Ana Maria Rey
Physical Review Research, 8, L032025 (2026)

26. *Reconfigurable dissipative entanglement between many spin ensembles: from robust quantum sensing to many-body state engineering*
Anjun Chu, Mikhail Mamaev, Martin Koppenhöfer, Ming Yuan, Aashish A. Clerk
Physical Review X, 16, 021047 (2026)

25. *Extracting information from a superradiant burst using simple measurements*
Federico Belliardo, **Anjun Chu**, Martin Koppenhöfer, Aashish A. Clerk
arXiv:2603.13130 (2026)

24. *Simulation of topological superconductors and their competing orders using photon-mediated interactions*
Anjun Chu, Joyce Kwan, Eric Yilun Song, Seth Hew Peng Chew, James K. Thompson, Ana Maria Rey
arXiv:2512.17889 (2025)

23. *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*
Minhui Zhu, Minyang Tian, Xiaocheng Yang, Tianci Zhou, Lifan Yuan, Penghao Zhu, Eli Chertkov, Shengyan Liu, Yufeng Du, Ziming Ji, Indranil Das, Junyi Cao, Yufeng Du, Jiabin Yu, Peixue Wu, Jinchen He, Yifan Su, Yikun Jiang, Yujie Zhang, Chang Liu, Ze-Min Huang, Weizhen Jia, Yunkai Wang, Farshid Jafarpour, Yong Zhao, Xinan Chen, Jessie Shelton, Aaron W. Young, John Bartolotta, Wenchao Xu, Yue Sun, **Anjun Chu**, Victor Colussi, Chris Akers, Nathan Brooks, Wenbo Fu, Jinchao Zhao, Marvin Qi, Anqi Mu, Yubo Yang, Allen Zang, Yang Lyu, Peizhi Mai, Christopher Wilson, Xuefei Guo, Juntai Zhou, Daniel Inafuku, Chi Xue, Luyu Gao, Ze Yang, Yaïr Hein, Yonatan Kahn, Kevin Zhou, Di Luo, John Drew Wilson, Jarrod T. Reilly, Dmytro Bandak, Ofir Press, Liang Yang, Xueying Wang, Hao Tong, Nicolas Chia, Eliu Huerta, Hao Peng
arXiv:2509.26574 (2025)

22. *Realization of three and four-body interactions between momentum states in a cavity through optical dressing*
Chengyi Luo, Haoqing Zhang, Chitose Maruko, Eliot A. Bohr, **Anjun Chu**, Ana Maria Rey, James K. Thompson
Science 390, 925 (2025)

21. *Solitons in arbitrary dimensions stabilized by photon-mediated interactions*
Haoqing Zhang, **Anjun Chu**, Chengyi Luo, James K. Thompson, Ana Maria Rey
Physical Review Letters 135, 173402 (2025)

20. *Atomic coherence of 2 minutes and instability of 1.5×10^{-18} at 1 s in a Wannier-Stark lattice clock*
Kyungtae Kim, Alexander Aeppli, William Warfield, **Anjun Chu**, Ana Maria Rey, Jun Ye
Physical Review Letters 135, 103601 (2025)

19. *Symmetry-protected topological optical lattice clock*
Tianrui Xu, **Anjun Chu**, Kyungtae Kim, James K. Thompson, Jun Ye, Tilman Esslinger, Ana Maria Rey
PRX Quantum 6, 030322 (2025)

18. *Time-resolved spectral gap spectroscopy in a quantum simulator of fermionic superfluidity inside an optical cavity*
Dylan J. Young, Eric Yilun Song, **Anjun Chu**, Diego Barberena, Zhijing Niu, Vera M. Schäfer, Robert J. Lewis-Swan, Ana Maria Rey, James K. Thompson
Physical Review Letters 134, 183404 (2025)

17. *Coherent evolution of superexchange interaction in seconds-long optical clock spectroscopy*
William R. Milner, Stefan Lannig, Mikhail Mamaev, Lingfeng Yan, **Anjun Chu**, Ben Lewis, Max N. Frankel, Ross B. Hutson, Ana Maria Rey, Jun Ye
Science 388, 503 (2025)

16. *A dissipation-induced superradiant transition in a strontium cavity-QED system*
Eric Yilun Song, Diego Barberena, Dylan J. Young, Edwin Chaparro, **Anjun Chu**, Sanaa Agarwal, Zhijing Niu, Jeremy T. Young, Ana Maria Rey, James K. Thompson
Science Advances 11, eadu5799 (2025)

15. *Hamiltonian engineering of collective XYZ spin models in an optical cavity*
Chengyi Luo, Haoqing Zhang, **Anjun Chu**, Chitose Maruko, Ana Maria Rey, James K. Thompson
Nature Physics 21, 916 (2025)

14. *Many-body gap protection of motional dephasing of an optical clock transition*
Zhijing Niu, Vera M. Schäfer, Haoqing Zhang, Cameron Wagner, Nathan R. Taylor, Dylan J. Young, Eric Yilun Song, **Anjun Chu**, Ana Maria Rey, James K. Thompson
Physical Review Letters 134, 113403 (2025)

13. *Exploring the dynamical interplay between mass-energy equivalence, interactions and entanglement in an optical lattice clock*
Anjun Chu, Victor J. Martínez-Lahuerta, Maya Miklos, Kyungtae Kim, Peter Zoller, Klemens Hammerer, Jun Ye, Ana Maria Rey
Physical Review Letters 134, 093201 (2025)

12. *Entangled matter-waves for quantum enhanced sensing*
John D. Wilson, Jarrod T. Reilly, Haoqing Zhang, Chengyi Luo, **Anjun Chu**, James K. Thompson, Ana Maria Rey, Murray J. Holland
Physical Review A 110, L041301 (2024)

11. *Trade-offs between unitary and measurement induced spin squeezing in cavity QED*
Diego Barberena, **Anjun Chu**, James K. Thompson, Ana Maria Rey
Physical Review Research 6, L032037 (2024)

10. *Momentum-exchange interactions in a Bragg atom interferometer suppress Doppler dephasing*
Chengyi Luo, Haoqing Zhang, Vanessa P. W. Koh, John D. Wilson, **Anjun Chu**, Murray J. Holland, Ana Maria Rey, James K. Thompson
Science 384, 551 (2024)

9. *Observing dynamical phases of BCS superconductors in a cavity QED simulator*
Dylan J. Young*, **Anjun Chu***, Eric Yilun Song, Diego Barberena, David Wellnitz, Zhijing Niu, Vera M. Schäfer, Robert J. Lewis-Swan, Ana Maria Rey, James K. Thompson
Nature 625, 679 (2024)

8. *Control and amplification of Bloch oscillations via photon-mediated interactions*
Haoqing Zhang, **Anjun Chu**, Chengyi Luo, James K. Thompson, Ana Maria Rey
Physical Review Research 5, L032039 (2023)

7. *Photon-mediated correlated hopping in a synthetic ladder*
   Anjun Chu, Asier Piñeiro Orioli, Diego Barberena, James K. Thompson, Ana Maria Rey
   Physical Review Research 5, L022034 (2023)

6. *Bosonic pair production and squeezing for optical phase measurements in long-lived dipoles coupled to a cavity*
   Bhuvanesh Sundar, Diego Barberena, Asier Piñeiro Orioli, **Anjun Chu**, James K. Thompson, Ana Maria Rey, Robert J. Lewis-Swan
   Physical Review Letters 130, 113202 (2023)

5. *Hamiltonian engineering of spin-orbit-coupled fermions in a Wannier-Stark optical lattice clock*
   Alexander Aeppli*, **Anjun Chu***, Tobias Bothwell*, Colin J. Kennedy, Dhruv Kedar, Peiru He, Ana Maria Rey, Jun Ye
   Science Advances 8, eadc9242 (2022)

4. *Quantum enhanced cavity QED interferometer with partially delocalized atoms in lattices*
   Anjun Chu, Peiru He, James K. Thompson, Ana Maria Rey
   Physical Review Letters 127, 210401 (2021)

3. *Observing the polarization of cosmic-ray muons in student physics laboratory*
   Jisi Xu, **Anjun Chu**, Zi Yuan, Xinliang Song, Tianle Cheng, Han Yi, Yijie Wang, Ying Chang, Zhao Zhang, Zhigang Xiao
   European Journal of Physics 42, 035603 (2021)

2. *Simulation of XXZ spin models using sideband transitions in trapped bosonic gases*
   Anjun Chu, Johannes Will, Jan Arlt, Carsten Klempt, Ana Maria Rey
   Physical Review Letters 125, 240504 (2020)

1. *Observation of dynamical quantum phase transitions in a spinor condensate*
   Haoxiang Yang, Tian Tian, Yanbin Yang, Liyuan Qiu, Haiyu Liang, **Anjun Chu**, Ceren B. Dağ, Yong Xu, Yingmei Liu, Luming Duan
   Physical Review A 100, 013622 (2019)

Invited Talks

6. *Reconfigurable dissipative stabilization of entanglement between many spin ensembles*
   Gordon Research Seminar (GRS) for Quantum Science (2026/07/26)

5. *Reconfigurable dissipative stabilization of entanglement between many spin ensembles: From robust quantum sensing to many-body state engineering*
   CQI Seminar (Online talk), Tsinghua University (2026/06/11)

4. *When light makes atoms talk: From quantum simulation to precision measurements*
   AMO/QI Seminar, University of California Berkeley (2026/02/04)

3. *Reconfigurable dissipative stabilization of entanglement between many spin ensembles: Robust quantum sensing and many-body state engineering*
   CTQM Theory Colloquium, University of Colorado Boulder (2025/08/22)

2. *Simulation of BCS superconductors in cavity QED systems*
   Boeing Quantum Creators Prize Symposium, University of Chicago (2023/11/14)

1. *Photon-mediated correlated hopping in a synthetic ladder*
   CTQM Theory Colloquium, University of Colorado Boulder (2022/10/21)

Teaching Experience

- **MENG 37200 Quantum Dissipation and Quantum Measurement, UChicago** *Spring 2026*
  Covering lectures when Prof. Aashish Clerk is traveling
  Graduate-level course introducing basic theoretical concepts and tools for open quantum systems and quantum measurements
- **PHYS 1120 General Physics 2, CU Boulder** *Spring 2019*
  Teaching assistant: Recitation class and grading
  Second semester of three-semester sequence for undergraduate-level general physics, covering electricity and magnetism, wave motion and optics
- **PHYS 1110 General Physics 1, CU Boulder** *Fall 2018*
  Teaching assistant: Recitation class and grading
  First semester of three-semester sequence for undergraduate-level general physics, covering kinematics, dynamics, momentum of particles and rigid bodies, work and energy, gravitation, and simple harmonic motion
- **PHYS 2170 Foundations of Modern Physics, CU Boulder** *Fall 2018*
  Teaching assistant: grading
  Third semester of three-semester sequence for undergraduate-level general physics, covering special relativity, quantum mechanics and atomic structure